

Reviewer Pack — Minimal Hodge Index Correlation with Communication Complexit...

Entry 03c6c412c5f0 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hodge Index Correlation with Communication Complexity Rank

Entry ID: 03c6c412c5f0 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 16:20:00 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Theory) **Field B** (complexity object): Communication Complexity

Statement:

For every CNF formula φ , the minimal Hodge index $h(\varphi)$ of its associated algebraic curve is linearly correlated with its communication complexity rank $r(\varphi)$, such that $h(\varphi) = \Theta(r(\varphi))$.

Rationale (proposer's reasoning):

Hodge theory provides a framework for studying algebraic varieties, including curves. If there exists a correspondence between the geometric properties of these varieties and the complexity of solving related computational problems, it could offer new insights into the nature of computation.

Taxonomy category: AlgebraicGeometryCommunicationComplexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 7e4a6cdd2573bf9c

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For every CNF formula, if the Pearson correlation coefficient between Hodge index and communication complexity rank is significant (p-value < 0.05) with a mean absolute difference in ranks ≤ 0.1 across all seeds, it supports the conjecture.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.95	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 2 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Hodge index" AND "communication complexity rank" - "algebraic geometry" AND CNF formula" - "minimal Hodge index" correlation WITH "communication complexity"

Top relevant hits considered: - [s2:e8b22f93b5f2611c0c5d621556fb1d5b02b7f532] Private Laticonic Oblivious Transfer with Preprocessing - [s2:10.58346/jisis.2024.i4.027] Lightweight Image Encryption Using Chacha20 and Serpent Algorithm

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(n):
            clause = [random.randint(1, n), -random.randint(1, n)]
            clauses.append(clause)
        return clauses

    def compute_communication_complexity(cnf):
        # Placeholder function; actual computation depends on the problem
        return len(cnf) * 2

    def compute_hodge_index(cnf):
        # Placeholder function; actual computation depends on the problem
        return len(cnf) / 2

    n = random.randint(5, 40)
    cnf = generate_cnf(n)
    comm_complexity_rank = compute_communication_complexity(cnf)
    hodge_index = compute_hodge_index(cnf)

    return {
        "metric_name": "communication_complexity_rank",
        "metric_value": comm_complexity_rank,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
```

```

        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 30)) # Default to first 30 primes if

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = (sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results)) ** 0.5
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in enumerate(results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

ication_complexity_rank', 'metric_value': 36, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 30, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 14, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 50, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 46, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 18, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 76, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 32, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 72, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 38, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
TRIAL: {'metric_name': 'communication_complexity_rank', 'metric_value': 34, 'instances_tested': 1, 'n_max': 18, 'conjecture_holds': True}
RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed=0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code uses placeholder functions for `compute_communication_complexity` and `compute_hodge_index`, which are not implemented according to the conjecture's mathematical definitions. This means that the empirical test is not measuring the correct metrics as stated in the conjecture.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test code uses placeholder functions for compute_communication_complexity and compute_hodge_index that are not implemented according to the conjec | next: Implement the correct mathematical definitions for compute_communication_complexity and compute_hodge_index as per the conjecture, then retest the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17518
2	preregistration	ollama_remote	glm4:latest	0	0	9297
3	novelty	ollama_remote	glm4:latest	0	0	8496
4	novelty	ollama_remote	glm4:latest	0	0	14666
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18020
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9410
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12930
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6476
9	critic	ollama_remote	glm4:latest	0	0	10738
10	judge	ollama_remote	glm4:latest	0	0	9161

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 116712 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/03c6c412c5f0.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/03c6c412c5f0.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/03c6c412c5f0.tar.gz (if generated)